

Savitribai Phule Pune University

LGNSCOE, Nashik — Department of Computer Engineering

B.E. Final Year Project

Project Synopsis

Academic Year: 2026–2027
Guide:

1 Group Information

Group ID

Group Members

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2 Project Title

Smart Kumbh Mela Safety & Assistance Platform Using Artificial Intelligence, IoT and Computer Vision

3 Project Option

Internal Project.

4 Internal Guide

5 Sponsorship and External Guide

Not applicable. This is an internal (non-sponsored) project, carried out entirely under the guidance of an internal faculty guide, with no external industry sponsorship or external guide involved.

6 Technical Keywords (As per ACM Keywords)

1. I. Computing Methodologies

(a) I.2 ARTIFICIAL INTELLIGENCE

i. I.2.10 Vision and Scene Understanding

A. Object recognition

B. Video analysis

(b) I.4 IMAGE PROCESSING AND COMPUTER VISION

i. I.4.8 Scene Analysis — Object recognition, Tracking

ii. I.4.9 Applications

(c) I.5 PATTERN RECOGNITION

i. I.5.4 Applications — Computer vision, Signal processing

2. C. Computer Systems Organization

(a) C.2 COMPUTER-COMMUNICATION NETWORKS

i. C.2.4 Distributed Systems

A. Client/server

B. Distributed applications

C. Cloud-based synchronization

3. H. Information Systems

(a) H.3 INFORMATION STORAGE AND RETRIEVAL

i. H.3.5 Online Information Services — Web-based services

(b) H.5 INFORMATION INTERFACES AND PRESENTATION

i. H.5.2 User Interfaces

4. K. Computing Milieux

(a) K.4 COMPUTERS AND SOCIETY

i. K.4.2 Social Issues — Public safety, Emergency management

7 Problem Statement

The Kumbh Mela in Nashik is among the largest human gatherings on Earth, drawing millions of devotees within a compressed window of time and space. This scale creates a chain of interconnected challenges: the absence of a centralized monitoring mechanism

makes it difficult for authorities to track crowd density in real time, which in turn leads to overcrowding at key ghats and junctions. Visitors, particularly children and the elderly, frequently get separated from their families with no fast, technology-driven way to reunite them. Medical emergencies arise unpredictably in dense crowds, and manual, radio-based coordination between volunteers, medical camps, and control rooms is too slow to guarantee timely response. Volunteers themselves lack a real-time view of tasks and their status, resulting in ad-hoc, inefficient task assignment. Without a unified digital system tying these functions together, an event that is manageable in principle can cascade into safety failures at scale.

There is, therefore, a need for a single, integrated, AI- and IoT-driven platform that brings crowd monitoring, lost-person recovery, emergency response, navigation, and volunteer coordination under one real-time system accessible to visitors, volunteers, and administrators alike.

8 Abstract

Mass religious gatherings such as the Kumbh Mela pose severe crowd-safety and coordination challenges due to the sheer scale of attendance concentrated in limited physical space and time. Existing crowd-management practices rely largely on manual observation and disconnected communication channels, which are inadequate for anticipating overcrowding, locating separated visitors, or responding quickly to medical emergencies. This project proposes a Smart Kumbh Mela Safety & Assistance Platform that unifies six interconnected modules under a shared, cloud-synchronized architecture: AI-based crowd monitoring using YOLOv8 and OpenCV to estimate real-time zone-wise crowd density and raise congestion alerts; a lost-person detection module combining QR-coded wristbands with face-recognition matching across camera feeds to reunite separated children, elderly visitors, and family members; an Emergency SOS module built on Flutter, Firebase, and the Google Maps API that lets a visitor summon the nearest help camp with a single tap; a smart navigation module that recalculates routes dynamically based on live crowd-density data rather than static maps alone; a volunteer-management module that lets administrators dispatch and track volunteers for lost-person cases, medical emergencies, and crowd-control tasks; and a centralized admin dashboard that consolidates live crowd status, active emergencies, lost-person reports, and volunteer availability on a single screen. The backend is built on Node.js and Express.js with a MySQL database, the admin interface in React, and the visitor/volunteer-facing app in Flutter, with JWT-based authentication securing all access. By replacing fragmented manual coordination with a single, real-time source of truth, the proposed system aims to reduce emergency response times, improve visibility for organizers, and materially improve visitor safety at one of the world’s largest public gatherings. The design is intentionally built on free and open-source components to keep it replicable for other mass-gathering events beyond the Kumbh Mela.

9 Goals and Objectives

1. Improve overall visitor safety throughout the duration of the event.
2. Reduce response time for medical emergencies and crowd-related incidents.

3. Provide event authorities a real-time, zone-wise view of crowd density.
4. Help families quickly locate lost or separated members, especially children and the elderly.
5. Guide visitors away from congested areas through live, density-aware route suggestions.
6. Enable volunteers to receive, accept, and complete assigned tasks efficiently through a dedicated interface.

10 Relevant Mathematics Associated with the Project

System Description

Input:

- Live video/CCTV feed frames from each monitored zone.
- Visitor registration data (photo, QR code, contact details) at enrolment.
- Real-time GPS coordinates from the visitor/volunteer mobile application.
- Volunteer status and task-acceptance events.

Output:

- Per-zone people count, density classification (Low / Medium / High / Critical), and heat-map visualization.
- Matched identity and location when a registered face is detected on any camera feed.
- Optimal, congestion-aware navigation route to a chosen destination.
- Volunteer-task assignment and live status updates on the admin dashboard.

Data Structures, Classes and Distributed-Processing Strategies:

- *Graphs* model the venue's paths and junctions, with nodes as waypoints/camps and edges as walkable paths, used for shortest-path route computation.
- *Priority queues* manage volunteer task dispatch (emergency severity as priority key) and SOS request handling.
- *Hash maps / vector indexes* store face-embedding vectors for $O(1)$ -average lookup during face-matching.
- *Spatial grids (zone partitioning)* divide the venue into monitored zones so that each camera feed can be processed independently.
- *Divide-and-conquer / distributed processing*: each zone's video stream is processed as an independent unit (map step) by a dedicated detection worker; zone-level counts are then aggregated (reduce step) into the global crowd-density view, allowing the crowd-monitoring module to scale horizontally as more camera feeds are added.

- *Classes/objects* include **Visitor**, **Volunteer**, **Emergency**, **Zone**, and **Task**, each mapped to its own database entity and API resource.

Functions — Objects, Morphisms, Overloading, Functional Relations:

- Detection function: $f_{\text{detect}} : \text{Frame} \rightarrow \{(b_i, c_i)\}$, mapping an image frame to a set of bounding boxes b_i with class confidence c_i (person class), via the YOLOv8 model.
- Density function: $D(z) = \frac{N(z)}{A(z)}$, where $N(z)$ is the detected people count in zone z and $A(z)$ is the zone's area, classified against threshold bands to yield the alert level.
- Face-similarity function: $\text{sim}(e_1, e_2) = \frac{e_1 \cdot e_2}{\|e_1\| \|e_2\|}$ (cosine similarity between two face-embedding vectors), with a match declared when $\text{sim}(e_1, e_2) \geq \tau$ for a tuned threshold τ .
- Route-cost function: $w(u, v) = d(u, v) \times \lambda(u, v)$, where $d(u, v)$ is the base physical distance between nodes u and v and $\lambda(u, v) \geq 1$ is a congestion multiplier derived from live density data; the navigation module then applies Dijkstra's / A* shortest-path algorithm over this dynamically weighted graph.
- These functions are overloaded across contexts (e.g., the detection function is reused identically for both crowd monitoring and lost-person face detection, differing only in the downstream classification head), illustrating functional reuse across modules that share the same underlying computer-vision pipeline.

Success Conditions:

- Crowd-density estimation matches ground-truth counts within an acceptable error margin at real-time (near-video-rate) throughput.
- Face-matching correctly reunites a registered lost person with an acceptably low false-positive rate.
- Navigation routes are recalculated and served within an interactive response time as density data updates.
- SOS requests are received, geo-located, and routed to the nearest volunteer/camp without failure.

Failure Conditions:

- Camera feed loss, poor lighting, or extreme occlusion causing missed detections.
- GPS signal unavailability preventing accurate SOS or navigation geolocation.
- Network/connectivity loss between mobile app, backend, and admin dashboard breaking real-time synchronization.
- False face-matches arising from visual similarity in extremely dense crowds.

11 Names of Conferences / Journals Where Papers Can Be Published

- IEEE International Conference on Computing, Communication and Networking Technologies (ICCCNT)
- IEEE International Conference on Advanced Computing and Communication Systems (ICACCS)
- ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems
- IEEE Region 10 Conference (TENCON)
- International Conference on Smart Cities, Automation & Intelligent Computing Systems
- Savitribai Phule Pune University (SPPU) sponsored Avishkar / student research conventions
- IIT Bombay Techfest Research Paper Symposium
- International Journal of Engineering Research & Technology (IJERT)
- IEEE Access (Journal)

12 Review of Conference / Journal Papers Supporting the Project Idea

1. **Intelligent Real-Time Crowd Density Estimation for Proactive Event Safety: A Machine Learning Approach**, Maharajpet, Sheela S., and Ananya V. Hegde, *Recent Research Reviews Journal*, 2025. Proposes a YOLOv8-plus-CSRNet pipeline that jointly detects and estimates crowd density from live CCTV/drone feeds, classifying zones into four density levels and exposing results through a real-time analytics dashboard – directly informing the crowd-monitoring module’s detection-plus-classification design.
2. **Dense-Stream YOLOv8n: A Lightweight Model for Real-Time Crowd Monitoring**, Research Square / Scientific Reports, 2025. Introduces a pruned, knowledge-distilled YOLOv8n variant that substantially raises frame-rate throughput while retaining detection accuracy in high-density, occluded scenes – relevant to keeping the platform’s per-zone detection pipeline real-time on modest hardware.
3. **Crowd Density Mapping and Anomaly Detection using YOLOv8 and DeepSORT**, R Discovery, 2025. Combines YOLOv8 detection with DeepSORT tracking to generate interactive density heat-maps and flag abnormal crowd behaviour, evaluated on benchmark datasets with strong precision and near-real-time latency – supports the heat-map and alert-generation design used in Module 1.

4. **An Enhanced Framework for Real-Time Dense Crowd Abnormal Behaviour Detection using YOLOv8**, Nasir et al., *Multimedia Tools and Applications* (Springer), 2025. Presents a Crowd Anomaly Detection Framework built on an improved YOLOv8 with Soft-NMS, validated on pilgrimage-scale datasets (Hajj) – a close analogue to the religious mass-gathering setting of the Kumbh Mela.
5. **Searchious: Locating Missing People using an Optimised Face Recognition Algorithm**, IEEE Conference Publication, 2021. Describes a face-recognition-based tool for tracing missing persons using a searchable database, reducing manual search effort – the core idea reused in the platform’s lost-person face-matching pipeline.
6. **Implementation of Machine Learning and KNN Algorithm for Finding Missing Person**, IEEE Conference Publication, 2022. Proposes a system where a guardian uploads a photo that is matched against a live database via face recognition, alerting both guardian and authorities on a match – structurally similar to the registration-to-alert flow in Module 2.
7. **Suhail: A Deep Learning-Based System for Identifying Missing People**, ResearchGate, 2023. A TensorFlow-based face-recognition system that lets families upload photos of missing persons and enables volunteer collaboration in the search process – supports the volunteer-assisted identification design used alongside face recognition.
8. **Efficient Face Recognition System for Identifying Lost People**, IJEAT, 2019. Uses a face-recognition pipeline to match uploaded photos of missing persons against camera-detected faces and alerts police/guardians with the match location – an early reference architecture for the QR-plus-face-recognition reunification flow adopted here.
9. **Managing Crowds with Technology: Cases of Hajj and Kumbh Mela**, Jambi et al., PMC, 2018. Surveys the use of RFID, wireless sensor networks, fog/cloud computing, and GPS for crowd management at Hajj and Kumbh Mela specifically, comparing their trade-offs for large periodic religious gatherings – provides direct domain grounding for the platform’s technology choices.
10. **A Smart Android Application for Kumbh Mela Management (BharatKumbh)**, IJPREMS, 2025. Proposes a centralized mobile platform integrating GPS navigation, QR-code-based crowd control, emergency response, and chatbot assistance specifically for the Nashik Kumbh Mela – closely parallels the overall system scope proposed in this project.
11. **Specifying the Effective Use of Artificial Intelligence for Crowd Management in Prayag Maha Kumbh Mela**, IJRAR, 2025. Examines AI-camera and AI-drone-based real-time crowd density analysis for planning and directing crowd flow at the Maha Kumbh Mela, reinforcing the case for AI-driven, camera-based monitoring at Kumbh-scale events.
12. **Smart Crowd Management Strategies for Kumbh Mela 2025**, IJSAT, 2025. Reviews IoT, AI, GIS, and mobile-application-based strategies for real-time monitoring, wearable tracking, and AI-powered surveillance at Kumbh Mela, framing technology adoption as central to safe mass-gathering management – supports the platform’s overall problem framing and objectives.

13 Plan of Project Execution

The project will be executed in phased sprints tracked through a project-management/planner tool (e.g., Trello or GitHub Projects), starting in August with system architecture and workflow design and proceeding through implementation, integration, and evaluation.

Phase	Timeline	Activities
Phase 1	August 2026	Requirement analysis; finalize system architecture and end-to-end workflow; define module boundaries and API contracts.
Phase 2	September 2026	Database design (Visitor, Volunteer, Emergency, Crowd/Zone tables); environment and repository setup; UI/UX wireframes for dashboard and mobile app.
Phase 3	October – November 2026	Development of Module 1 (AI Crowd Monitoring: YOLOv8 + OpenCV pipeline) and Module 2 (Lost Person Detection: QR registration + face recognition).
Phase 4	December 2026 – January 2027	Development of Module 3 (Emergency SOS: Flutter + Firebase + Google Maps API) and Module 4 (Smart Navigation with density-aware routing).
Phase 5	January – February 2027	Development of Module 5 (Volunteer Management) and Module 6 (Admin Dashboard); integration of all modules over the shared backend/cloud sync layer.
Phase 6	February – March 2027	System-level testing, bug fixing, performance and load testing, security hardening (JWT-based auth).
Phase 7	March – April 2027	Final documentation, report preparation, and viva/presentation readiness.